PROJECT REPORT

Job Market Intelligence Platform

An End-to-End Data Analytics & Machine Learning Project

Prepared By	J. Charan Reddy
Project Type	End-to-End Data Analytics Portfolio Project
Domain	Job Market Analysis — US Data Roles
Date	March 2026
Tech Stack	Python, MySQL, Streamlit, Scikit-learn
Deployment	Streamlit Cloud (Live Application)

Data Analyst & Data Scientist Portfolio | March 2026

1. Executive Summary

This report documents the complete development of the Job Market Intelligence Platform, an end-to-end data analytics project built to analyze the US data job market in 2026. The platform was designed and built by J. Charan Reddy as a portfolio project demonstrating full-stack data science capabilities — from live API data collection to machine learning model deployment.

The project spans four phases: data collection and storage, exploratory data analysis and NLP, statistical hypothesis testing, and machine learning with deployment. A total of 929 live job listings were collected across 6 data roles using the JSearch REST API and stored in a MySQL database. The final deliverable is a live interactive web application deployed on Streamlit Cloud featuring salary prediction, skill intelligence, and market analysis.

Key Highlights

Total Jobs Collected	929 live job listings across 6 data roles
Data Source	JSearch REST API via RapidAPI (real-time, not Kaggle)
Database	MySQL — 929 records, 13 columns, 13 advanced SQL queries
NLP Analysis	spaCy + NLTK VADER on 929 job descriptions
ML Model	Gradient Boosting — MAE $34,885 | R^2 0.429 | GridSearchCV tuned
Statistical Tests	T-test & Mann-Whitney U — 2 hypotheses validated
Deployment	Live Streamlit app with 8 pages and user activity tracking

2. Project Overview

2.1 Objectives

The primary objective of this project was to build a comprehensive, production-grade Job Market Intelligence Platform that demonstrates end-to-end data science skills including data engineering, SQL, NLP, statistical testing, machine learning, and deployment.

- Collect real-time job market data via REST API for 6 data roles in the US
- Store and query data using MySQL with advanced SQL techniques
- Perform exploratory data analysis and NLP on job descriptions
- Conduct statistical hypothesis testing to validate salary assumptions
- Build and deploy a machine learning salary predictor
- Deploy a live interactive dashboard accessible via public URL

2.2 Scope

The project covers the United States job market for the following 6 data roles: Data Analyst, Data Scientist, Business Analyst, Data Engineer, Machine Learning Engineer, and BI Analyst. All data was collected in March 2026 via the JSearch API, which aggregates listings from LinkedIn, Indeed, Glassdoor, and other major job boards.

2.3 Tech Stack

Category	Technologies
Data Collection	Python, requests, JSearch API, RapidAPI
Data Storage	MySQL, SQLAlchemy, CSV
Data Processing	Pandas, NumPy, Pathlib
NLP	spaCy (en_core_web_sm), NLTK VADER, WordCloud, regex
Statistics	SciPy (ttest_ind, mannwhitneyu)
Machine Learning	Scikit-learn, XGBoost, Gradient Boosting, GridSearchCV, Pickle
Visualization	Plotly Express, Matplotlib, Seaborn
Deployment	Streamlit, Streamlit Cloud, GitHub

3. Phase 1 — Data Collection & Storage

3.1 API Configuration

Job data was collected using the JSearch API hosted on RapidAPI. The API was queried with 20 pages per role, yielding approximately 10 jobs per page. Authentication was handled via the x-rapidapi-key header.

- **https://jsearch.p.rapidapi.com/search** API Endpoint:
- **Data Analyst, Data Scientist, Business Analyst, Data Engineer, Machine Learning Engineer, BI Analyst** Roles Queried:
- **20 pages (approximately 10 results per page)** Pages per Role:
- **982 records fetched across all 6 roles** Total Raw Jobs:

3.2 Data Collection Results

Role	Jobs Fetched	After Cleaning	% of Total
Data Analyst	129	123	13.2%
Data Scientist	166	158	17.0%
Business Analyst	176	167	18.0%
Data Engineer	155	147	15.8%
Machine Learning Engineer	180	171	18.4%
BI Analyst	176	163	17.5%
TOTAL	**982**	**929**	**100%**

3.3 Data Cleaning

Raw data was cleaned using Pandas method chaining. The following transformations were applied:

- Selected 13 relevant columns from the raw API response
- Imputed missing employment type values with 'Unknown'
- Imputed missing salary period values with 'Not Disclosed'
- Calculated avg_salary as the mean of job_min_salary and job_max_salary
- Fixed 'Bi Analyst' label to 'BI Analyst' using regex pattern matching
- Final cleaned dataset: 929 rows x 13 columns, zero null values

3.4 MySQL Database

The cleaned dataset was loaded into a MySQL database using SQLAlchemy. The database serves as the primary storage and query layer for the project.

- **job_market** Database Name:
- **jobs** Table Name:
- **929** Total Records Loaded:
- **mysql+mysqlconnector://root@localhost:3306/job_market** Connection:

3.5 SQL Analysis — 13 Advanced Queries

Thirteen SQL queries were written to extract business insights from the database, including advanced techniques such as window functions, CTEs (Common Table Expressions), RANK() OVER (PARTITION BY), NTILE(), and conditional aggregation.

#	Query Description	Key Finding
1	Top hiring companies by job count	Capital One — 28 listings
2	Average salary by role	ML Engineer highest at $69K avg
3	Remote vs onsite job count	97.4% onsite, 2.6% remote
4	Top cities by listing count	Washington DC — 83 listings
5	Salary ranking using RANK() OVER	Apple highest at $238K avg
6	Role distribution percentage	All 6 roles nearly equal (13-18%)
7	Top paying cities per role (CTE)	San Francisco for ML Engineer ($415K)
8	Salary decile using NTILE(10)	Top 10% earn above $200K
9	Jobs by employment type	Full-time dominates at 85%+
10	Companies with multiple roles	Walmart hires across all roles
11	Remote salary comparison	Remote avg $99K vs onsite $129K
12	Running total of jobs (SUM OVER)	East Coast accounts for 60%+ jobs
13	Salary band classification (CASE WHEN)	45% jobs in $50K-$100K band

4. Phase 2 — Exploratory Data Analysis & NLP

4.1 EDA Charts

Eight interactive visualizations were created using Plotly Express with the plotly_dark theme. All charts were saved as PNG files to the charts/ directory for use in the Streamlit dashboard.

#	Chart	Key Insight
1	Salary by Role (Boxplot)	ML Engineer highest, Data Analyst lowest
2	Top Companies (Bar)	Capital One #1, Virtual Vocations #2
3	Remote vs Onsite (Donut)	97.4% onsite, only 2.6% remote
4	Top Cities (Bar)	Washington DC dominates with 83 jobs
5	Role x State Heatmap	Virginia and DC lead across all roles
6	Salary by Employment Type	Full-time highest, contractor variable
7	Correlation Heatmap	Salary & has_salary: 0.8 correlation
8	Top Skills (Bar)	Communication #1, then SQL and Python

4.2 NLP Pipeline

Natural Language Processing was applied to 929 job descriptions using spaCy (en_core_web_sm) and NLTK. The NLP pipeline extracted top in-demand skills, bigrams, and sentiment scores from raw job description text.

Skill Extraction

A custom skill vocabulary of 30 technical and soft skills was matched against cleaned job descriptions using regular expressions. The top skills identified were:

- Communication — most mentioned skill across all roles
- Excel — top non-programming technical skill
- Machine Learning — top advanced technical skill
- Python and SQL — most demanded programming skills
- Scala — highest paying technical skill correlation

Bigram Analysis

The top bigrams extracted from 929 job descriptions using NLTK revealed the most common two-word phrases in the US data job market:

- 'machine learning' — 1,000+ mentions, most common bigram

- 'bachelors degree' — second most common, showing education requirements
- 'data science' and 'data analysis' — core domain terms

Sentiment Analysis

Sentiment scores were computed using NLTK's VADER (Valence Aware Dictionary and sEntiment Reasoner) model, which is specifically designed for social and professional text.

- Over 800 job descriptions scored near 1.0 compound sentiment score
- All 6 roles clustered near positive sentiment (compound > 0.9)
- Data Scientist role showed one negative outlier in the distribution
- Conclusion: Employers use highly positive, aspirational language in job postings

5. Phase 3 — Statistical Hypothesis Testing

Statistical hypothesis testing was conducted on 324 jobs that had salary data available (Avg Salary > 0). Both parametric (T-test) and non-parametric (Mann-Whitney U) tests were applied at a significance level of α = 0.05.

5.1 Sample Breakdown

Role	Jobs with Salary Data
Data Scientist	66
Machine Learning Engineer	60
Business Analyst	55
BI Analyst	53
Data Engineer	49
Data Analyst	41
Total	**324**

5.2 Hypothesis 1 — Data Scientist vs Data Analyst Salary

Metric	Data Scientist	Data Analyst
Sample Size	66	41
Mean Salary	$166,500	$104,200
Median Salary	$160,000	$100,000
T-Statistic	6.735	—
P-Value (T-test)	0.0000	—
P-Value (Mann-Whitney)	0.0000	—
Conclusion	**REJECT Null Hypothesis**	**Significant Difference**

Interpretation: With a p-value of 0.0000 (well below α = 0.05), we reject the null hypothesis. There is strong statistical evidence that Data Scientists earn significantly more than Data Analysts in the US market. The median salary gap is approximately $60,000 per year.

5.3 Hypothesis 2 — Remote vs Onsite Salary

Metric	Remote Jobs	Onsite Jobs
Sample Size	6	318
Mean Salary	$99,970	$129,120
T-Statistic	-0.913	—
P-Value (T-test)	0.3618	—
P-Value (Mann-Whitney)	0.3022	—
Conclusion	**FAIL TO REJECT**	**No Significant Difference**

Interpretation: With p = 0.3618, we fail to reject the null hypothesis. The difference in salary between remote and onsite jobs is not statistically significant. This is partly due to the very small remote sample (n=6). The finding confirms that salary should not be the primary decision factor when choosing between remote and onsite roles.

6. Phase 3 — Machine Learning & Salary Predictor

6.1 Dataset Preparation

The ML model was trained on the 324 jobs that had salary data. Features were label-encoded and the dataset was split into 80% training (259 records) and 20% testing (65 records) using a fixed random state of 42 for reproducibility.

- **Role, Job City, Job State, Job Is Remote, Job Employment Type** Features Used:
- **Avg Salary (continuous)** Target Variable:
- **LabelEncoder applied to all categorical features** Encoding:
- **80% / 20% (random_state=42)** Train/Test Split:

6.2 Model Comparison

Three tree-based regression models were trained and compared using the same train/test split to identify the best performer before hyperparameter tuning.

Model	MAE	RMSE	R^2 Score
Random Forest	$37,695	$53,252	0.325
Gradient Boosting	$37,332	$51,474	0.369
XGBoost	$41,128	$59,221	0.165
Winner	**Gradient Boosting**	**Best MAE & R^2**	**Selected for tuning**

6.3 Hyperparameter Tuning — GridSearchCV

GridSearchCV with 5-fold cross validation was applied to the Gradient Boosting model. A grid of 16 parameter combinations was evaluated, totalling 80 fits.

- **[100, 200]** n_estimators:
- **[3, 5]** max_depth:
- **[0.05, 0.1]** learning_rate:
- **[2, 5]** min_samples_split:
- **learning_rate=0.05, max_depth=3, min_samples_split=2, n_estimators=100** Best Parameters Found:
- **$45,385** Best CV MAE:

6.4 Final Model Performance

Metric	Before Tuning	After Tuning
MAE	$37,332	$34,885
RMSE	$51,474	$48,953
R^2 Score	0.369	0.429
CV MAE (5-fold)	—	$46,508 ± $13,582

6.5 Feature Importance

Feature importance was extracted from the tuned Gradient Boosting model to understand which input variables have the greatest influence on salary prediction.

Rank	Feature	Importance	Insight
1	Role	0.45	What you do matters most
2	Job City	0.21	Location is key
3	Job Employment Type	0.22	Contract vs full-time matters
4	Job State	0.11	State has moderate influence
5	Job Is Remote	0.01	Remote barely affects salary

7. Phase 4 — Streamlit Dashboard & Deployment

7.1 Application Overview

The final phase involved building and deploying an interactive web application using Streamlit. The app serves as the front-end interface for all project outputs — EDA charts, NLP insights, statistical test results, and the ML salary predictor.

7.2 Application Pages

#	Page	Description
1	**Overview**	Hero section, 5 KPI cards, role distribution chart, role summary table, market recommendations
2	**Market Analysis**	Interactive filters (role/state/work type), charts, recommendations, full market data table
3	**NLP — Skill Intelligence**	Skill bar chart, demand % bars, role heatmap, sentiment charts, skills priority table, learning recommendations
4	**Statistical Testing**	H1 and H2 results with metrics, box plots, test summary table, salary stats table, career recommendations
5	**Salary Predictor**	Input form, prediction card, personalized recommendations, role comparison table, top companies chart
6	**Raw Data Explorer**	Filter controls, full data table, CSV download, summary statistics, charts
7	**User Activity & Extracts**	Activity log, prediction history, filter selections, usage charts, CSV extraction buttons
8	**About This Project**	Project methodology, tech stack, key findings, resume bullet points

7.3 User Activity Tracking

The application automatically tracks all user interactions and saves them to CSV files in the data/tracking/ directory. Three tracking files are maintained:

- **— records every page visit and action with timestamp** activity_log.csv
- **— records every salary prediction with inputs and outputs** salary_predictions.csv
- **— records every filter change across all pages** filter_selections.csv

8. Key Findings & Insights

8.1 Market Findings

- Washington DC is the #1 hiring city for data roles with 83 listings, driven by government and defense contracts
- Capital One is the most active recruiter with 28 job postings across multiple data roles
- 97.4% of all data jobs are onsite — remote roles represent only 2.6% of the market in 2026
- The East Coast (DC, New York, Virginia) accounts for over 60% of all data job listings

8.2 Salary Findings

- Machine Learning Engineers command the highest salaries with a maximum of $675,000 observed
- Apple offers the highest average salary at $238,000 among all companies in the dataset
- Data Scientists earn significantly more than Data Analysts — median gap of $60,000 ($p=0.0000$)
- San Francisco offers the highest ML Engineer salaries averaging $415,000
- New York leads for Data Engineer salaries averaging $253,000

8.3 Skills Findings

- Communication is the most mentioned skill across all 6 data roles
- 'Machine Learning' is the most common two-word phrase in 1,000+ job descriptions
- Python, SQL, and Excel are the top 3 technical skills demanded across all roles
- Cloud skills (AWS, Azure, GCP) are increasingly required especially for Data Engineers
- 86%+ of all job descriptions use positive sentiment language with VADER scores above 0.9

8.4 ML Model Findings

- Role is the single most important salary predictor with 45% feature importance
- Job City and Employment Type together account for 43% of salary variance
- Remote vs onsite status barely influences salary prediction (1% feature importance)
- Gradient Boosting outperforms Random Forest and XGBoost on this dataset
- Final model achieves $R^2 = 0.429$ — explains 43% of salary variance on limited data

9. Conclusions & Career Recommendations

9.1 Project Conclusions

This project successfully demonstrates an end-to-end data science workflow on a real-world problem. Starting from raw API data and ending with a deployed interactive web application, every stage of the data science lifecycle was implemented and documented.

The platform provides actionable insights for data professionals looking to navigate the US job market. Key conclusions include the dominance of onsite work arrangements, the significant salary premium for Data Scientists over Data Analysts, and the critical importance of role and location in determining compensation.

9.2 Career Recommendations

- Data Analyst to Data Scientist transition offers the highest salary ROI — the gap is statistically proven
- Target Washington DC, New York, and San Francisco for maximum job exposure and salary potential
- Master Python, SQL, and Machine Learning as the core non-negotiable skills across all roles
- Do not rely on remote work as a primary job search strategy — only 2.6% of roles are remote
- For ML Engineering, San Francisco companies pay significantly above national averages

9.3 Limitations

- Salary data was only available for 324 of 929 jobs (34.9%) — predictions have higher uncertainty
- Remote sample size (n=6) is too small for robust statistical conclusions in H2
- Data was collected at a single point in time — market conditions change
- JSearch API may not cover all job boards — some niche roles may be underrepresented

9.4 Future Enhancements

- Collect salary data from additional sources to improve model accuracy
- Add time-series analysis to track job market trends over multiple months
- Implement NLP-based job description similarity search for personalized job matching

- Expand to international job markets (UK, Canada, Australia)
- Add automated weekly API refresh to keep data current